



RGB-D + instruction
 ``put the mug on the plate``

VLM / perception grounding

local, per-object:
 Referenced(x)
 Protected(x), Moving(x)
 positions: open-vocab
 detection + depth

*neural: answers
 local predicates*

FOL rule $\rightarrow$ Hazard(x)

$\text{Prot}(x) \vee \text{Mov}(x) \vee$
 $(\neg \text{Ment}(x) \wedge \text{NearPath}(x))$

3-regime routing

Avoid (keep-out)
 Disengage (hazard = dest.)
 Refuse (unsafe instr.)

*symbolic: makes the decision
 execute certified 5-step prefix; re-observe and replan*

$K = 8$ chunks

noise-space steering

select / tilt / ascend the noise
 of the frozen flow VLA;
 actions stay on-manifold

checked certificate

DCBF chain re-verified on
 the executed prefix
 (mechanism-independent)